

GROUP ROLEX

Recommendation System

Vanessa Faller & Sasha Granelli



Data Science & Machine Learning – December 15th, 2024

Introduction

- 📖 **Recommendation system** : Personalize user experiences and simplify content discovery.
- 📖 **Application** : widely used across industries, from streaming platforms (Netflix, Spotify) to e-commerce (Amazon) and social media (YouTube, TikTok).
- 📖 **Goal** : Build a recommendation system for a university library to help students and researchers discover books.
- 📖 **Techniques** : Collaborative filtering and text embedding.

Objectives

- 📖 **Develop a Functional Recommendation System** : Data preprocessing, training algorithms and optimizing results to ensure accuracy.
- 📖 **Address Personalization Challenge** : Predict books based on user interests and library catalog metadata.
- 📖 **Improve User Experience** : User-friendly system, enhance engagement with the library's platform.
- 📖 **Showcase Role of Recommendation Systems** : Demonstrate how advanced algorithms solve information overload in large catalogs.

Methodology(1)

Data Analysis

- **Data Preprocessing** : Understanding of the datasets, cleaning and standardizing data for accuracy and consistency.
- **Datasets** : User interactions, book metadata, and sample submissions.
- **Format Compliance** : Ensured compatibility with Kaggle competition requirements.

Methodology(2)

Model Exploration

- Collaborative Filtering Approach

- **User-item matrix** each row represented a user, each column a book, and the values indicated binary interactions.
- **Cosine similarity** : Calculated user-user and item-item similarity matrices.
- **Prediction Methods** : Item-based and user-based prediction methods.
- **Evaluation** : Precision@K and Recall@K metrics to assess model performance.
- **Initial Results** : Precision score improved from **0.1241** to **0.1452** after retraining on the full dataset.

Methodology(3)

Advanced Model Improvements

- **User-item Matrix built from interactions timestamps** : the timestamps indirectly prioritize items with multiple interactions (0.1452 $\rightarrow$ 0.1592).
- **Time Decay Model** : applies an exponentially decreasing weighting based on the age of interactions to prioritize recent ones.
- **Hybrid Approach** : Combined user-to-user and item-to-item similarities, adjusting weights assigned to each of them and fine tuning them to maximize performance.
- **Improvement** : Precision score increased from **0.1452** to **0.1694**.

Methodology(4)

Best Model

- **Textual Metadata Integration** : Used “DistilRoBERTa”, “multi-qa-MiniLM-L6-cos-v1”, and “all-mpnet-base-v2” pre-trained models for semantic embeddings.
- **Result** : Minor improvement in prediction accuracy (**0.1698**), highlighting potential for further optimization.

Challenges

- 📖 **Combining Approaches** : Difficulty balancing collaborative filtering with semantic embeddings.
- 📖 **Data Limitations** : Lack of explicit user interactions reduced data richness.
- 📖 **Outcome** : Improved predictions, but gains were less significant than expected.

A photograph of a grand, ornate library interior. The room features high ceilings with intricate wooden carvings and a balcony with a decorative railing. The walls are lined with tall, dark wooden bookshelves filled with books. A prominent red staircase with wooden railings leads up to the balcony. The floor is covered with stacks of books and papers, suggesting a place of active research or study. The lighting is warm and focused, highlighting the architectural details and the vast collection of books.

University Library Application

Conclusion

- 📖 **Model Evolution** : Transitioned from collaborative filtering to a hybrid approach with time decay and textual embeddings, improving accuracy ($0.1452 \rightarrow 0.1698$).
- 📖 **Limitations** : Modest improvements from textual embeddings, indicating room for optimization.
- 📖 **Key Takeaway** : Recommendation systems simplify access to large catalogs, meet user needs, and highlight the role of data science in creating personalized experiences.



Thank you for your attention

Visit our [GitHub](#) for more details !